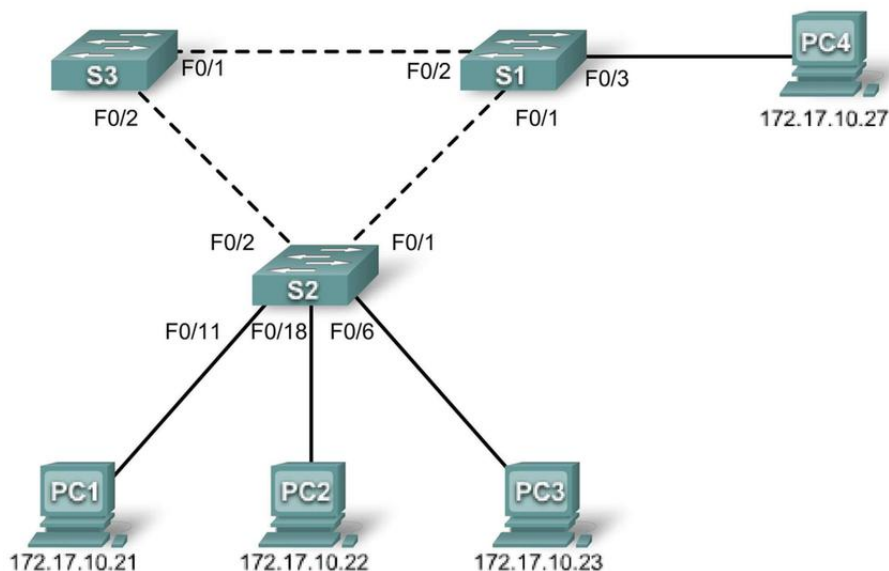




Guia 5– Redundancia L2

Objectivos:

- Executar a configuração básica de spanning tree nos switches
- Observar e explicar o comportamento por defeito do Protocolo de Spanning Tree (STP, 802.1D)
- Observar a resposta a uma mudança à topologia de spanning tree
- Agregação de Links através de LACP



1. Efetue a montagem da rede de acordo com o diagrama anterior. Para implementar os switches S1, S2 e S3 utilize L3SW.

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	172.17.10.1	255.255.255.0	N/A
S2	VLAN 1	172.17.10.2	255.255.255.0	N/A
S3	VLAN 1	172.17.10.3	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.254
PC2	NIC	172.17.10.22	255.255.255.0	172.17.10.254
PC3	NIC	172.17.10.23	255.255.255.0	172.17.10.254
PC4	NIC	172.17.10.27	255.255.255.0	172.17.10.254

2. Configure os PCs de forma a passarem a ter os endereços IP identificados na tabela anterior.

3. Atribua o *hostname* dos switches de acordo com o nome definido na tabela anterior.

Cofinanciado por:





```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z. Switch(config)#hostname S1
S1(config-line)#end
%SYS-5-CONFIG_I: Configured from console by console
S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

4. Desligue todos os portos usando o comando shutdown.

```
S1(config)#interface range fa0/1-24
S1(config-if-range)#shutdown

S2(config)#interface range fa0/1-24
S2(config-if-range)#shutdown

S3(config)#interface range fa0/1-24
S3(config-if-range)#shutdown
```

5. Religue as interfaces em utilização dos swichs.

```
S1(config)#interface range fa0/1-3
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown

S2(config)#interface range fa0/1-3
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown

S1(config)#interface range fa0/1-2
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown
```

6. Religue as interfaces em utilização dos swichs.



S1#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
Address 0019.068d.6980 This is the MAC address of the root switch

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0019.068d.6980
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.3	P2p
Fa0/2	Desg	FWD	19	128.4	P2p
Fa0/3	Desg	FWD	19	128.5	P2p

S2#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
Address 0019.068d.6980
Cost 19
Port 1 (FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 001b.0c68.2080
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/6	Desg	FWD	19	128.6	P2p
Fa0/11	Desg	FWD	19	128.11	P2p
Fa0/18	Desg	FWD	19	128.18	P2p

S3#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
Address 0019.068d.6980
Cost 19
Port 1 (FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 001b.5303.1700
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Cofinanciado por:





Interface	Role	Sts	Cost	Prio.	Nbr	Type
Fa0/1	Root	FWD	19	128.1		P2p
Fa0/2	Altn	BLK	19	128.2		P2p

7. Examine a saída dos switchs. O Bridge ID, armazenado no BPDU do spanning tree, consiste na bridge priority, na extensão do system ID, e o endereço MAC. A combinação ou adição da bridge priority e a system ID extension são conhecidas como bridge ID priority. A system ID extension é sempre o número da VLAN. Por exemplo, o system ID extension para a VLAN 100 é 100. Usando o bridge priority padrão de 32768, a prioridade da ID bridge ID priority for VLAN 100 seria 32868 (32768 + 100).

8. Examine a saída dos switchs. O Bridge ID, armazenado no BPDU do spanning tree, consiste na bridge priority, na extensão do system ID, e o endereço MAC. Com base na saída, qual é o bridge ID priority for switches S1, S2, and S3 on VLAN 1?

- a. S1 _____
- b. S2 _____
- c. S3 _____

Qual o root switch para spanning tree da VLAN 1? _____

No S1, which as portas do spanning tree que estão no estado blocking state switch root?

No S3, which as portas do spanning tree que estão no estado blocking state? _____

Como o STP elege o switch root? _____

Como as prioridades da bridge são todas iguais, o que mais o switch usa para determinar a raiz?

9. Observe a alteração da topologia 802.1D STP. Active o debug dos eventos spanning tree.

- 1. S1#debug spanning-tree events Spanning Tree event debugging is on
- 2. S2#debug spanning-tree events Spanning Tree event debugging is on
- 3. S3#debug spanning-tree events Spanning Tree event debugging is on

10. Desligue utilizando o comando shutdown uma das interfaces.

S1(config)#interface fa0/1 S1(config-if)#shutdown

11. Analise a saída do S2 e S3



Step 3: Record the debug output from S2 and S3

```
S2#  
1w2d: STP: VLAN0001 we are the spanning tree root  
S2#  
1w2d: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,  
changed state to down  
1w2d: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down  
S2#  
1w2d: STP: VLAN0001 heard root 32769-0019.068d.6980 on Fa0/2  
1w2d:      supersedes 32769-001b.0c68.2080  
1w2d: STP: VLAN0001 new root is 32769, 0019.068d.6980 on port Fa0/2, cost 38  
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/2  
  
S3#  
1w2d: STP: VLAN0001 heard root 32769-001b.0c68.2080 on Fa0/2  
1w2d: STP: VLAN0001 Fa0/2 -> listening  
S3#  
1w2d: STP: VLAN0001 Topology Change rcvd on Fa0/2  
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/1  
S3#  
1w2d: STP: VLAN0001 Fa0/2 -> learning  
S3#  
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/1  
1w2d: STP: VLAN0001 Fa0/2 -> forwarding
```

12. Quando o link do S2 que está conectado ao switch raiz cai, qual é a sua conclusão inicial sobre a raiz da árvore de abrangência? _____

13. Uma vez que S2 recebe nova informação sobre Fa0 / 2, que nova conclusão ela tira? _____

14. A porta Fa0 / 2 no S3 estava anteriormente em um estado de bloqueio antes do link entre S2 e S1 cair. Por quais estados ele passa como resultado da mudança de topologia? _____

15. Examine o que mudou na topologia da árvore de abrangência usando o comando show spanning-tree



S2#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
 Address 0019.068d.6980
 Cost 38
 Port 2 (FastEthernet0/2)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 001b.0c68.2080
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p
Fa0/6	Desg	FWD	19	128.6	P2p
Fa0/11	Desg	FWD	19	128.11	P2p
Fa0/18	Desg	FWD	19	128.18	P2p

S3#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
 Address 0019.068d.6980
 Cost 19
 Port 1 (FastEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

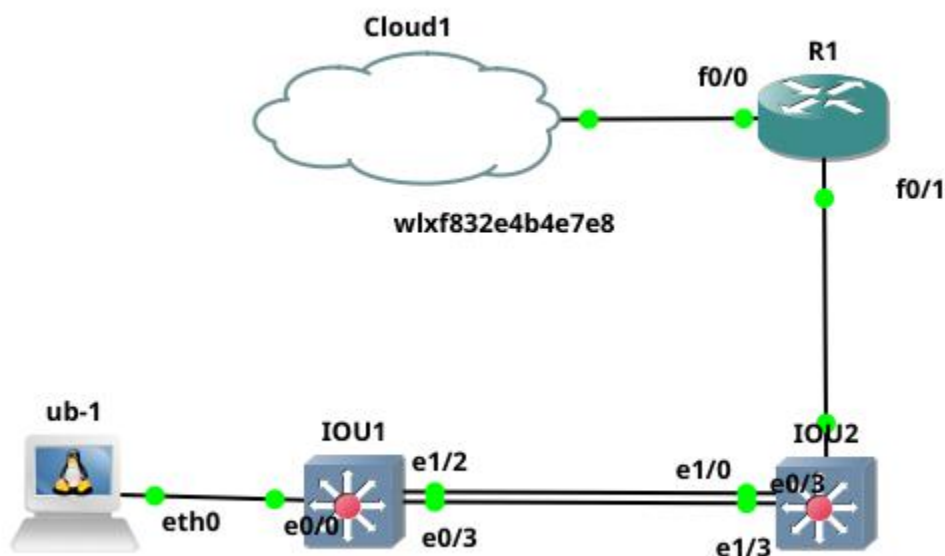
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 001b.5303.1700
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

16. O que mudou no modo como o S2 encaminha o tráfego?

17. O que mudou na forma como o S3 encaminha o tráfego?

Link Aggregation



18. Crie a rede da figura anterior configurando:

- O router para ter acesso ao exterior através do object cloud
- Ter a rede 10.0.0.0/8 na rede interna
- Configure NAT de forma a que a rede interna tenha acesso ao exterior através da porta f0/0 do router

19. Configure agregação dos links que ligam os switches:

..no SW IOU1

```
IOU1(config)#interface e1/2
```

```
IOU1(config-if)#channel-group 1 mode active
```

```
Creating a port-channel interface Port-channel 1
```

```
IOU1(config-if)#interface e0/3
```

```
IOU1(config-if)#channel-group 1 mode active
```

```
IOU1(config-if)#exit
```

```
IOU1(config)#
```

..no SW IOU2

```
IOU2#
```

```
IOU2#conf terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
IOU2(config)#interface e1/0
```

```
IOU2(config-if)#channel-group 1 mode active
```

```
Creating a port-channel interface Port-channel 1
```

Cofinanciado por:





```
IOU2(config-if)#interface e1/0  
IOU2(config-if)#interface e1/3  
IOU2(config-if)#channel-group 1 mode active  
IOU2(config-if)#exit  
IOU2(config)#
```

20. Teste a conectividade entre o PC e o router.